Global C-MAPSS RUL Study

Train on every FD001–FD004 training trajectory; evaluate the locked model on every official FD001–FD004 test set

Prepared 21 October 2026 | CPU-bounded local execution | 709 training engines | 707 official test engines

Protocol

The four training files were concatenated after remapping engine identifiers. Three HistGradientBoosting candidates were selected using only combined-training data: three subset-stratified engine-grouped folds and fixed 10/25/50/75/100/125-cycle pseudo-censored endpoints. The condition-normalized candidates fit six operating-condition clusters and sensor moments on development rows in each fold. Official test labels were read only after the validation choice was fixed.

Global validation selection

Candidate	Cap	Condition norm.	RMSE	MAE	R2	PHM08 mean
global_hgb_cap100_condition_normalized	100	yes	14.101	9.730	0.878	2.532
global_hgb_cap125_condition_normalized	125	yes	11.800	8.178	0.914	2.561
global_hgb_cap125_raw_sensor_control	125	no	17.169	12.526	0.819	7.546

Selected before official testing: **global_hgb_cap100_condition_normalized**. It is ranked by mean PHM08, followed by RMSE. The cap and normalizer are then refit on all 709 training engines before endpoint prediction.

Locked official-test results

Test subset	Engines	RMSE	MAE	R2	PHM08 mean	Bias
FD001	100	16.292	11.608	0.846	3.164	-7.688
FD002	259	31.128	19.183	0.665	61.477	-14.767
FD003	100	17.273	11.982	0.826	3.785	-6.686
FD004	248	33.391	21.785	0.625	62.574	-17.451

Pooled result

Across all 707 official test engines: RMSE **28.737**, MAE **18.006**, R2 **0.683**, and PHM08 mean **45.454**. This aggregate is descriptive; the per-subset rows remain the primary result because fault-mode and operating-condition coverage differs across FD001–FD004.

Interpretation and next step

This is the requested all-training-data model, evaluated consistently on every official test file. It measures performance after exposure to both single- and multi-condition data and both one- and two-fault training regimes. It should be compared with the earlier FD004-only transfer study as a separate experiment, because that study intentionally withheld FD001–FD003 training data. The saved endpoint predictions support further error analysis, calibration review, or a lightweight edge-model distillation experiment.

Reproducibility

Runner: run_all_subsets_global_study.py. The complete validation comparison, per-subset endpoint predictions, global model artifact, split inventories, and machine-readable metrics are stored in reports/all_subsets_global and artifacts/all_subsets_global.